Method-2

```
def factorial (n):
    if (n==0):
        return 1
    else:
        return n * factorial (n-1)
n= int (input ("Enter number : "))
c= factorial (n)
print (c)
```

Class and object in python

→ Python is an object oriented programming language.
→ Almost everything in python is an object with its properties and methods.

Creating a class and object

To create a class, use the keyword class.

Syntax : class class_name :
properties / methods

For ex -

```
class Myclass :
    x = 5
```
class

```
p1 = myclass()
print (p1.x)
```
object

The __init__() function

__init__() function is used to assign value to object

properties or other operations that are necessary to do when the object is being created.

The __init__() function is called automatically everytime the class is being used to create a new object.

Ques- WAP to create a class named person use the __init__() function to assign value for name and age.

Answer-
```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age
p1 = person ("Yashika", 18)
print (p1.name)
print (p1.age).
```

Attributes and Methods

→ Object can also contain methods.
→ Methods in object are functions that belongs to the object.

Ques- WAP to insert a function that prints a greeting and execute it on the object.

Answer-
```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age
    def myfunc(self):
        print ("Hello my name is" self.name)
p1 = Person ("Yashika", 18)
```

```
p1.myfunc()
Hello, my name is Yashika
```

Modify object properties

```
p1.age = 20
print (p1.age)
```

Delete object properties

```
del p1.age
```

Delete object

```
del p1
```

Data modelling

Data modelling in python is the general process by which this programming language organizes everything internally and it treats and process data.

The data model is the building block of python

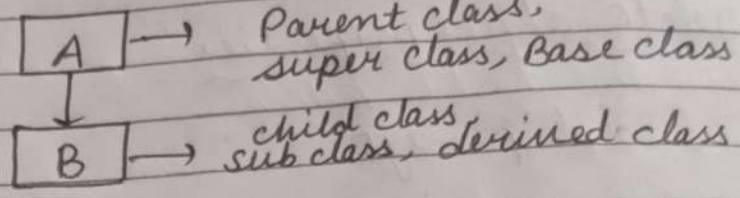
Internally Data model has its design and code of block for its implementation.

The process of creating data modules using the syntax and environment of the python programming language is called Data Modelling python.

Python Inheritance

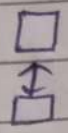
Inheritance is the capability of one class to acquire the properties of another class.

A new class is created by inheriting features of another class.

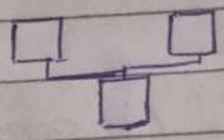


Types of inheritance

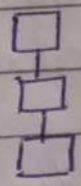
→ Single



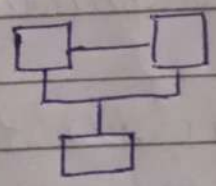
→ Multiple



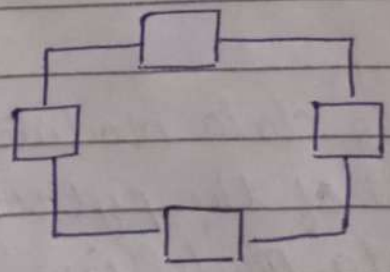
→ Multilevel



→ Heerarchial



→ Hybrid



Syntax : class parent_class :
    # body of parent class
class child_class (parent_class):
    # inherits the parent class
    # body of child class.

Example : class Person :
    def __init__ (self, name, age):
        self.name = name
        self.age = age
    def display_info (self):
        print ("Display Information:")
        print (f "Name : {self.name}")
        print (f "Age : {self.age}")

Output
Display information :
Name : John
Age : 35
Display information
Name : Alice
Age : 20
Stuid : 5001
course : B.Tech

class student (Person):
    def_init_ (self, name, age, st_id
    course)
    super ()__init__(name, age)
    self. st_id = st_id
    Self. course = course
    def display_info (self):
        Super () . display_info ()
        print (f "st_id {self.st_id}"
        print (f" course {self. course
Pers = Person ("John", 35)
Pers . display_info()
Stud = student ("Alice", 20,
    "5001", "B.Tech")
Stud. display_info ()